Monotonicity

Mind map

Monotonic function

- Only ↑ or only ↓ in domain
- Continuity not needed

At a point

- $f(a-h) < f(a) < f(a+h)$
- $f'(a) > 0$ sufficient

In an interval

- $x_1 > x_2 \Rightarrow f(x_1) > f(x_2)$
- Non-decreasing: $\geq$

Derivative test

- $f' \geq 0$ (isolated zeros)
- Sign scheme of f'
- $f \circ g$: same $\Rightarrow\uparrow$

Critical points

- $f' = 0$ or DNE
- Stationary: $f' = 0$

Always increasing

- $p \geq \max g$
- Cubic: $b^2 \leq 3ac$

Greatest / least value

- Ends + critical points
- a^b vs b^a: $\frac{\ln x}{x}$

Proving inequality

- $h = f - g$, $h(a) \geq 0$, $h' > 0$
- $f(A) > f(B) \iff A > B$ ($f \uparrow$)

Rolle's theorem

- $f(a) = f(b) \Rightarrow f'(c) = 0$
- Root of f' between roots of f
- $e^{kx}f$, $\frac{f}{x}$

LMVT

- $f'(c) = \frac{f(b)-f(a)}{b-a}$
- Tangent $\parallel$ chord
- $m(b-a) \leq \Delta f \leq M(b-a)$

Number of roots

- Sign changes on monotonic pieces
- $f' > 0$: one root

Plotting graph

- Domain, limits, f' sign
- Cubic, Q/Q

Asymptotes

- $x = a$: $f \to \infty$
- $y = \lim\limits_{x\to\infty} f$
- $y = mx + b$ by division